

STRUISBAAI, South Africa

WMO: 689210

Lat: 34.8003S

Lon: 20.0569E

Elev: 10

StdP: 14.69

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.8	44.0	37.5	33.0	46.2	39.1	35.1	48.2	28.6	61.9	26.9	60.5	5.2	290	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.0	77.4	69.2	76.2	68.5	75.0	67.8	71.8	75.3	70.7	74.3	69.7	73.5	12.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.6	112.7	74.2	69.3	107.5	73.4	68.2	103.6	72.7	35.6	75.3	34.6	74.5	33.8	73.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.8	22.3	36.7	82.2	2.8	2.7	34.7	84.1	33.0	85.6	31.5	87.1	29.4	89.1		
(5)			35.8	72.4	1.6	1.9	34.6	73.8	33.7	74.9	32.7	76.0	31.5	77.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	70.2	70.4	67.9	63.5	59.6	55.8	54.8	55.5	58.2	61.6	64.4	68.4	(6)
(7)		DBStd	6.43	3.01	2.62	3.08	3.40	3.58	3.36	3.66	3.41	3.42	3.31	2.94	2.99	(7)
(8)		HDD50	7	0	0	0	0	0	1	4	2	0	0	0	0	(8)
(9)		HDD65	1498	3	1	9	66	169	275	315	294	205	112	43	6	(9)
(10)		CDD50	4571	627	572	556	404	299	176	154	173	246	360	433	571	(10)
(11)		CDD65	586	165	153	100	20	3	0	0	0	1	6	26	112	(11)
(12)		CDH74	522	205	132	61	7	4	0	1	0	5	8	3	96	(12)
(13)		CDH80	12	2	1	3	1	0	0	0	0	0	2	0	3	(13)
(14)	Wind	WSAvg	10.5	11.4	11.1	10.9	10.2	9.3	8.9	9.5	9.7	10.5	11.5	11.6	11.1	(14)
(15)	Precipitation	PrecAvg	17.7	0.9	0.7	1.1	1.9	2.0	2.2	2.4	1.9	1.2	1.5	1.1	0.7	(15)
(16)		PrecMax	23.1	5.0	2.5	5.4	10.0	5.7	5.5	5.5	5.1	2.8	4.4	3.6	4.1	(16)
(17)		PrecMin	10.3	0.0	0.1	0.2	0.4	0.5	0.7	0.8	0.6	0.3	0.2	0.2	0.1	(17)
(18)		PrecStd	3.3	1.2	0.5	1.2	1.9	1.0	1.1	1.2	1.0	0.6	1.2	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.8	78.5	77.5	74.2	72.0	69.5	68.9	68.6	71.5	72.8	73.8	78.6	(19)
(20)			MCWB	71.6	69.7	69.6	64.7	60.7	58.6	58.7	57.8	60.4	62.1	65.5	68.7	(20)
(21)		2%	DB	77.6	76.7	75.6	71.3	68.9	65.6	65.4	65.0	67.0	69.6	71.9	76.3	(21)
(22)			MCWB	69.8	69.2	68.7	65.2	61.2	57.7	57.7	57.2	59.5	62.5	64.6	68.0	(22)
(23)		5%	DB	76.1	75.6	74.1	69.6	67.0	63.6	62.9	63.0	64.9	67.9	70.6	74.8	(23)
(24)			MCWB	68.7	68.5	68.0	64.6	61.0	57.6	57.0	56.6	58.9	61.7	63.6	67.0	(24)
(25)		10%	DB	74.7	74.5	72.7	68.2	65.3	62.0	61.1	61.2	63.4	66.4	69.2	73.3	(25)
(26)			MCWB	67.8	67.8	67.2	63.9	60.4	57.1	56.4	56.2	58.0	60.8	62.5	66.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	72.7	72.1	69.2	65.7	61.2	60.3	61.0	63.8	66.1	70.0	71.9	(27)
(28)			MCDWB	77.9	75.8	74.5	70.4	67.4	64.4	64.1	64.1	68.0	71.1	71.9	75.3	(28)
(29)		2%	WB	72.0	71.3	70.8	67.4	63.7	60.2	59.2	59.6	62.0	64.6	66.9	70.2	(29)
(30)			MCDWB	76.1	74.7	73.8	69.5	66.1	63.2	63.1	62.6	65.9	68.1	70.4	74.1	(30)
(31)		5%	WB	70.7	70.3	69.6	66.3	62.5	59.1	58.2	58.3	60.6	63.4	65.1	69.0	(31)
(32)			MCDWB	74.8	74.0	72.9	68.5	65.5	62.3	62.1	61.6	64.0	66.7	68.9	73.3	(32)
(33)		10%	WB	69.3	69.4	68.3	65.1	61.4	58.0	57.2	57.2	59.3	62.2	63.8	67.9	(33)
(34)			MCDWB	73.6	73.4	71.6	67.8	64.8	61.3	61.0	60.9	62.9	65.6	68.0	72.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.0	8.5	8.9	10.9	13.1	12.7	12.0	10.1	8.6	8.7	8.8	(35)
(36)			MCDBR	9.6	9.1	10.4	10.5	13.8	16.0	15.5	14.6	12.1	10.3	10.1	10.5	(36)
(37)			MCWBR	5.4	5.2	5.9	6.3	8.0	9.9	10.0	8.9	7.0	5.7	5.7	5.7	(37)
(38)		5% WB	MCDWB	8.4	7.4	7.6	7.7	10.1	12.5	13.6	11.5	10.8	8.2	8.0	8.6	(38)
(39)			MCWBR	5.3	4.9	4.9	5.5	6.9	9.1	10.0	8.1	7.0	5.3	5.2	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.362	0.365	0.353	0.335	0.328	0.321	0.321	0.340	0.352	0.369	0.355	0.362	(40)	
(41)		taud	2.507	2.508	2.539	2.574	2.574	2.569	2.560	2.499	2.446	2.421	2.494	2.496	(41)	
(42)		Ebn at Noon	309	301	293	281	265	258	264	274	288	296	309	310	(42)	
(43)		Edh at Noon	36	35	31	27	23	22	23	28	34	37	36	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2378	2118	1708	1264	921	780	857	1108	1516	1909	2278	2451	(44)	
(45)		RadStd	82	82	90	57	49	40	39	50	71	109	89	108	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200

Nomenclature: See separate page